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# 1.4 Weather and Climate

Eduqas GCSE Geography (C/4) — Component 1

## 1. Weather vs Climate

**Weather:** The short-term atmospheric conditions at a specific place and time — including temperature, rainfall, wind speed, humidity, cloud cover, and air pressure. Weather can change from hour to hour.

**Climate:** The *average* weather conditions of a place over a **long period of time** (typically 30 years). Climate describes the expected pattern of weather for a region rather than day-to-day variation.

### UK Climate Characteristics — Temperate Maritime

The UK has a **temperate maritime climate**, shaped primarily by its proximity to the Atlantic Ocean and the warming North Atlantic Drift current.

| Season | Temperature | Rainfall | Key influence |
|---|---|---|---|
| Winter | Mild (rarely below −5°C in lowlands) | Wet; frequent frontal rain and depressions | Atlantic Ocean stores summer heat and releases it slowly; North Atlantic Drift |
| Summer | Warm (rarely above 28°C in most areas) | Drier, though convectional rain possible | Longer days increase solar radiation; Atlantic limits extremes |
| Autumn/Spring | Transitional — no sharp boundaries | Variable; changeable | Interaction of air masses |

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Key characteristics of the UK climate:

- No extreme temperatures — cool summers and mild winters (compared to same latitude inland)
- Rainfall distributed throughout the year (no dry season)
- Western areas receive more rainfall than eastern areas (prevailing south-westerly winds)
- Highly variable, changeable day-to-day weather
- Influenced by several different **air masses** (Polar Maritime, Tropical Maritime, Polar Continental, Tropical Continental)

**Exam tip:** Don't confuse weather and climate. Weather = short-term, specific. Climate = long-term, average. A common exam question asks you to "distinguish between weather and climate" — always include both the time-scale difference and the idea of averages vs day-to-day conditions.

## 2. Factors Affecting UK Climate

| Factor | Effect on UK Climate | Example |
|---|---|---|
| **Latitude** | The UK (50–61°N) receives less concentrated solar radiation than tropical regions because the Sun is lower in the sky. The same amount of solar energy is spread over a larger surface area. Higher latitudes are cooler. | Scotland is colder than southern England because it is at higher latitude (more northerly) |
| **Altitude** | Temperature decreases with height at an average rate of **6.5°C per 1,000 m** (the environmental lapse rate). High ground is colder and receives more precipitation. Snow more likely on summits. | Cairngorm summit (1,245 m) is on average ~8°C colder than nearby Aviemore in the valley; over 100 days of snow cover/year |
| **Distance from the sea (continentality)** | The sea heats and cools more slowly than the land (*maritime effect*). Areas near the sea have smaller temperature ranges (mild winters, cool summers). Inland areas have a *continental climate* — hotter summers, colder winters, less rainfall. | London (inland) is hotter in summer and colder in winter than Plymouth (coastal). Eastern Europe has extremes not seen in W England. |

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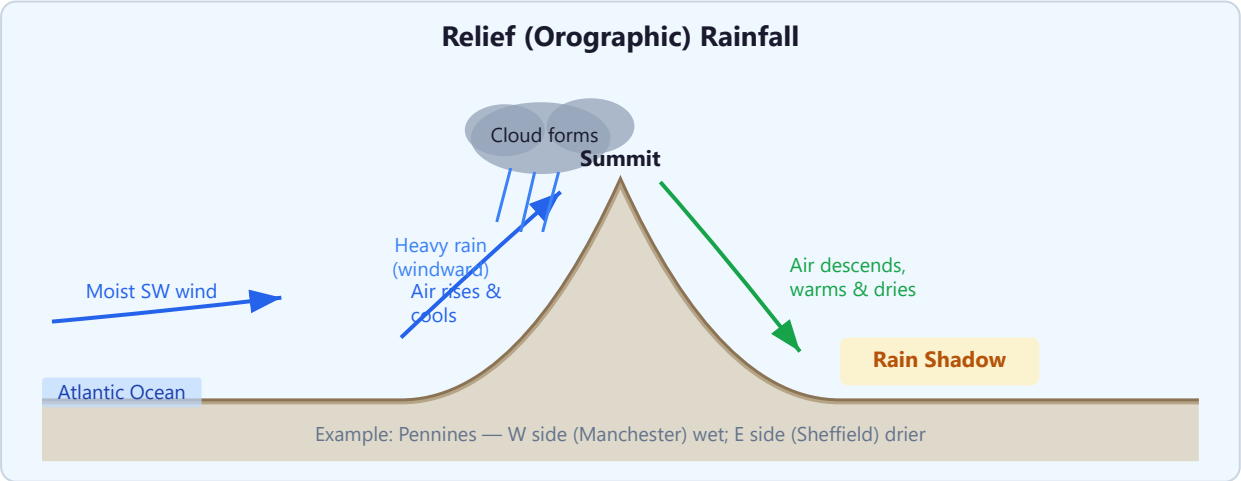
| | | |
|---|---|---|
| **Prevailing winds** | UK's prevailing winds come from the **south-west**, bringing warm, moist air from the Atlantic Ocean. This gives the west coast more rainfall. Easterly winds from continental Europe bring cold, dry conditions in winter and hot, dry air in summer. | Cardiff receives c. 1,150 mm rain/year; London receives c. 590 mm/year (sheltered from south-westerly winds by Wales) |
| **Ocean currents (North Atlantic Drift)** | The North Atlantic Drift (NAD) — an extension of the Gulf Stream — flows north-eastward past the British Isles, bringing warm tropical water. This keeps the UK significantly warmer than other places at the same latitude (e.g. Newfoundland, Canada). | Stornoway, Scotland (58°N) has January average of 5°C; Labrador, Canada (same latitude) averages −15°C in January |

## 3. Relief Rainfall

**Relief (orographic) rainfall:** Rainfall caused when moist air is forced to rise over a mountain barrier (orographic uplift). As the air rises, it cools (at the dry adiabatic lapse rate — 9.8°C/1,000 m until saturation; then at the saturated adiabatic lapse rate — ~6°C/1,000 m). Once cooled to the dew point, water vapour condenses into clouds and falls as rain on the *windward* side. On the *leeward* side (the rain shadow), descending air warms and dries out, producing lower rainfall.

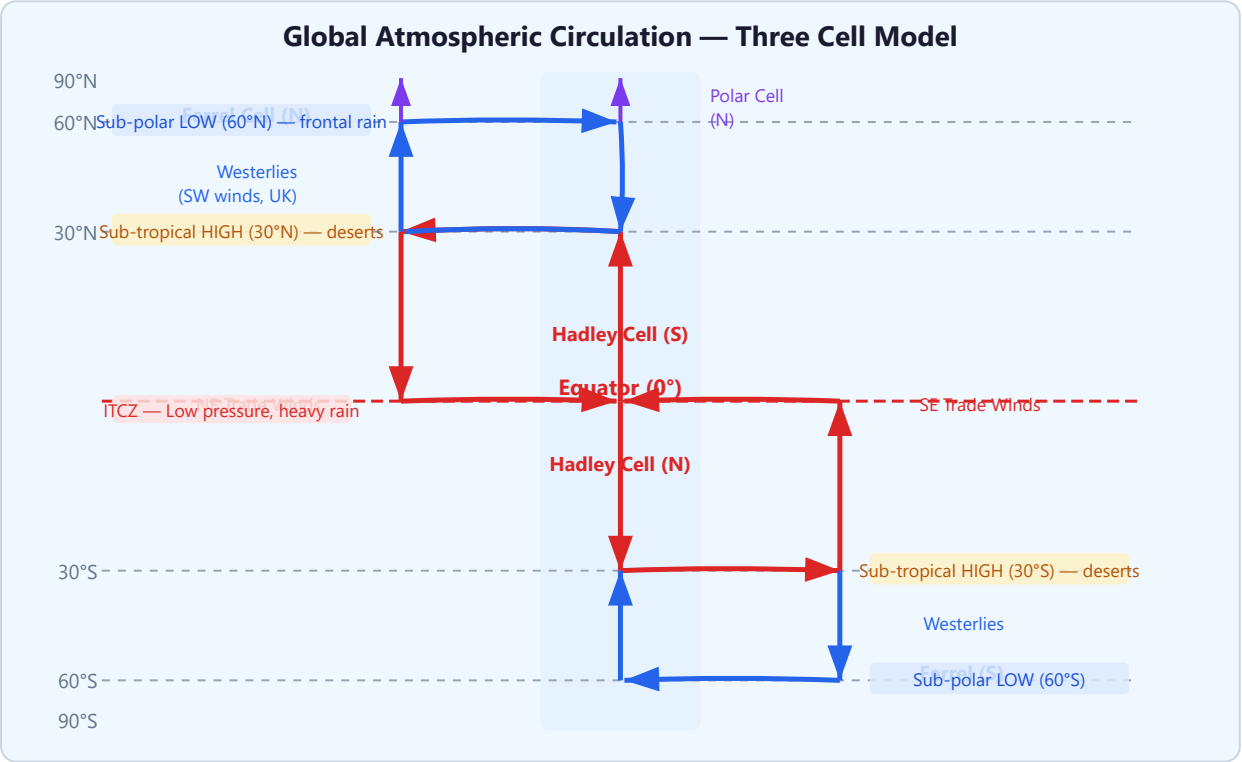



**Named example — the Pennines:** The Pennine hills run north–south through northern England. Prevailing south-westerly winds bring moist Atlantic air that rises over the Pennines. Manchester (west side) receives c. 800 mm of rain/year; Sheffield (east side, rain shadow) receives c. 650 mm/year. The difference is even

more pronounced in the higher Lake District fells — Seathwaite in the Lake District averages over 3,000 mm/year (one of England's wettest places), while the Vale of York to the east averages c. 600 mm/year.

## 4. Global Atmospheric Circulation

**Global atmospheric circulation:** The large-scale movement of air around the Earth driven by unequal heating. The equator receives more solar radiation than the poles, creating temperature and pressure differences. This drives a system of convection cells (Hadley, Ferrel, and Polar cells) that redistribute heat energy around the globe and create the major wind belts and climate zones.




| Pressure Belt / Zone | Latitude | Conditions | Climate result |
|---|---|---|---|
| **ITCZ** (Inter-Tropical Convergence Zone) | 0° (equator) | Rising air, intense heating, low pressure | Equatorial climate: hot, very wet all year; tropical rainforest |
| Sub-tropical high pressure | ~30°N/S | Descending air, high pressure, dry | Hot desert climates (Sahara, Arabian, Australian deserts) |

| | | | |
|---|---|---|---|
| Sub-polar low pressure | ~60°N/S | Cold polar air meets warm tropical air; fronts; rising air | Temperate climate with frontal rainfall (UK, NW Europe) |
| Polar high pressure | 90°N/S | Descending cold air, high pressure, very dry | Polar/tundra climate: very cold, low precipitation (mainly snow) |

## 5. Depressions

**Depression (mid-latitude cyclone):** An area of low atmospheric pressure formed when cold polar air meets warm tropical air. The boundary between these air masses is called the **polar front**. Warm air rises over cold air, creating low pressure at the surface. Depressions bring unsettled, cloudy, wet, and windy weather to the UK. They move broadly west to east across the UK, driven by the jet stream.

### Structure of a Depression

| Feature | Description |
|---|---|
| **Warm front** | Leading edge of the warm air mass, shown on maps as a line with semicircles. Warm air rises gently over cold air ahead of it — produces wide cloud cover and steady, prolonged rain (or drizzle). Passes before the warm sector arrives. |
| **Warm sector** | Area of warm air between the warm and cold fronts. Relatively mild, drier, and clearer spells (though some low cloud possible). |
| **Cold front** | Trailing edge where cold air undercuts and lifts the warm air rapidly — shown on maps as a line with triangles. Produces brief but intense rainfall, heavy showers, thunderstorms, then cold, showery conditions as the front passes. |
| **Occluded front** | As the faster cold front catches up with the warm front, the warm sector is lifted off the ground, forming an occluded front. The depression is weakening at this stage. |

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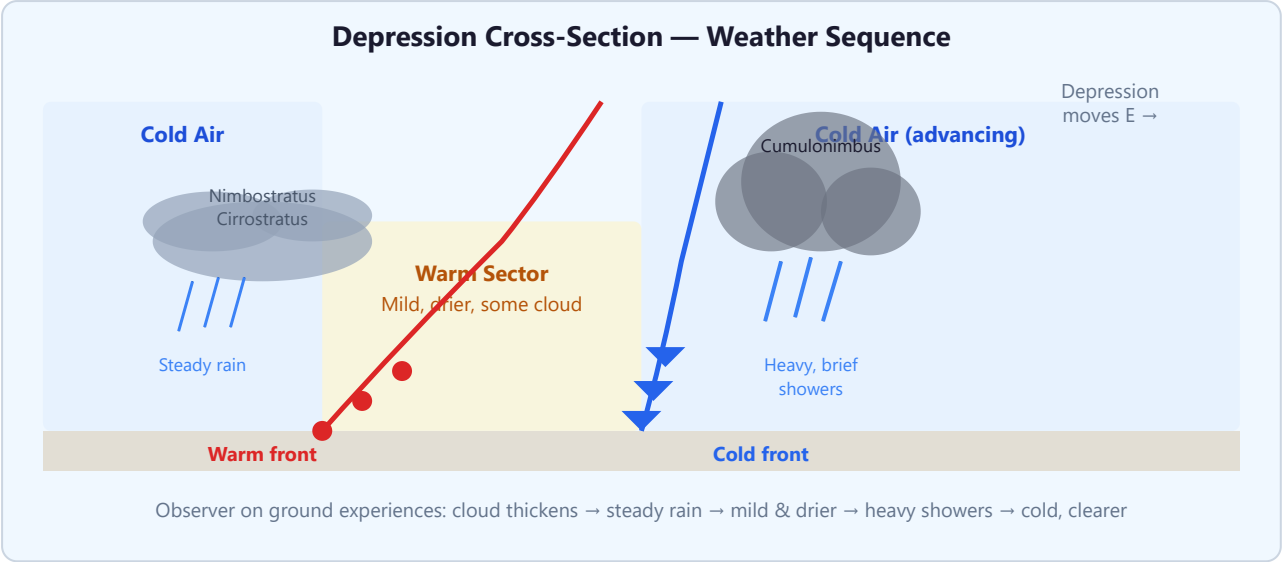


**Weather sequence as a depression passes over the UK (west to east):**

1. **Before warm front:** Cirrus clouds high in sky, then thickening cloud (cirrostratus, altostratus, nimbostratus). Pressure falling. Steady, prolonged rain begins.
2. **In warm sector:** Warmer temperatures. Rain eases. Possible drizzle; low cloud. Wind direction changes (veers from SW to S).
3. **Cold front passes:** Heavy, brief rain or thunderstorms. Winds strengthen and veer to NW. Temperature drops sharply.
4. **After cold front:** Cold, bright, showery conditions. Cumulus/cumulonimbus clouds. Good visibility between showers. Pressure rising.

## 6. Anticyclones

**Anticyclone:** An area of **high atmospheric pressure** where air descends from upper levels toward the surface, then diverges outward. Descending air is warmed by compression, which suppresses cloud formation. Anticyclones bring settled, dry, calm weather. They rotate clockwise in the Northern Hemisphere. They can persist for days or weeks, causing blocking weather patterns.

| Feature | Summer Anticyclone | Winter Anticyclone |
|---|---|---|
| Temperature | Hot, sunny (can reach 30°C+ in SE England) | Very cold (frost common; temperatures below 0°C at night) |

| | | |
|---|---|---|
| Precipitation | Dry; drought risk with prolonged anticyclones | Dry; any moisture forms fog or frost rather than rain |
| Cloud cover | Clear skies; intense sunshine | Clear at night (radiation fog); possible low stratus by day |
| Wind | Calm, light winds | Calm; cold temperatures persist because winds don't bring Atlantic warmth |
| Hazards | Heatwave, drought, UV risk, hosepipe bans | Black ice, freezing fog, hypothermia risk for elderly |
| Air quality | Poor in urban areas — pollutants trapped (temperature inversions possible) | Smog possible; poor visibility in fog |

## Depression vs Anticyclone Comparison

| Feature | Depression | Anticyclone |
|---|---|---|
| Pressure | Low (<1000 mb) | High (>1013 mb) |
| Air movement | Air converges and rises | Air descends and diverges |
| Wind speed | Strong; isobars close together | Light or calm; isobars widely spaced |
| Wind direction (N. Hemisphere) | Anticlockwise (cyclonic) | Clockwise (anticyclonic) |
| Cloud / precipitation | Heavy cloud; prolonged rain or showers | Clear skies; dry |
| Duration | Days (pass through quickly) | Days to weeks (blocking high) |
| Temperature | Variable; warm sector, then colder | Hot in summer; very cold in winter |

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**Exam tip:** A common 4-mark question asks you to describe the weather associated with either a depression or anticyclone. Use the table above and refer to temperature, precipitation, cloud cover, and wind. Always specify summer or winter for anticyclones as they produce very different conditions.

## 7. UK Extreme Weather Events

### Case Study: Summer Heatwaves

**2003 European Heatwave:** France and southern UK experienced temperatures 6–7°C above average for over two weeks. UK recorded 38.5°C (Brogdale, Kent). Caused by a persistent anticyclone blocking Atlantic systems. Over 70,000 excess deaths across Europe; 2,000+ in UK. Crops failed; wildfires; power stations shut as river water temperatures rose too high for cooling.

**2018 UK Heatwave:** Prolonged high-pressure system from June to August. Temperatures reached 35.3°C (Faversham, Kent). UK's joint-hottest summer on record at the time. Drought conditions: rivers at record lows; hosepipe bans; moorland fires in Greater Manchester (Saddleworth Moor).

**2022 UK Heatwave:** On 19 July 2022, the UK recorded its highest ever temperature: **40.3°C** at Coningsby, Lincolnshire. Caused by a plume of hot air from North Africa channelled north by an anticyclone. Wildfires in London suburbs; rail lines buckled; Luton Airport tarmac melted; schools and businesses closed. Linked to climate change — Met Office stated such temperatures would have been virtually impossible without human-caused climate change.

### Case Study: 2013–14 Somerset Levels Flooding

Winter 2013–14 was the UK's wettest in 248 years of records. A series of Atlantic depressions brought relentless storms. The Somerset Levels — a low-lying floodplain in SW England — were inundated from late December 2013 to March 2014.

- 65 sq km of farmland flooded; water depths up to 1 m; 17 million tonnes of water on the Levels
- Villages including Moorland and Muchelney were cut off for weeks by road
- 150 properties flooded; 1,000 livestock evacuated
- Estimated cost of damage: >£10 million; economic disruption to farms >£7 million

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- Controversy over the Environment Agency's decision to reduce dredging of rivers Tone and Parrett in prior years
- Long-term response: £100 million Somerset Levels Flood Action Plan; 8 km of rivers dredged; upgraded pumping stations; raised flood banks

**Exam tip:** When answering extreme weather questions, always include: specific dates, specific statistics (temperatures, cost, areas affected), social impacts AND economic impacts, and — for higher marks — link to the cause (e.g. blocking anticyclone, exceptional run of Atlantic depressions).

## 8. Global Climate Zones

| Climate Zone | Location | Temperature | Precipitation | Example |
|---|---|---|---|---|
| **Equatorial** | 0–5° N/S (equator) | Hot all year: 25–28°C; no seasons | Very heavy: 2,000–3,000 mm/yr; rain every afternoon | Amazon Basin; Congo Basin; Borneo |
| **Tropical (savanna)** | 5–20° N/S | Hot all year: 20–32°C; wet season cooler | Strong wet and dry seasons: 750–1,500 mm (seasonal) | Nigeria; Kenya; northern Australia |
| **Arid / Desert** | 20–30° N/S (sub-tropical) | Extreme range: 40–50°C day; near 0°C night | Very low: <250 mm/yr; unreliable | Sahara; Arabian Desert; Atacama |
| **Mediterranean** | 30–40° N/S (W coasts) | Hot dry summers (25–35°C); mild wet winters (8–15°C) | 500–800 mm; mostly in winter | Spain; Italy; California; SW Australia |
| **Temperate (maritime)** | 40–65° N/S (W coasts) | Mild: 0–22°C; no extremes | Regular rainfall all year: 600–1,500 mm | UK; France; NW USA; New Zealand |

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|---|---|---|---|---|
| **Continental** | 40–65°N (inland) | Extreme range: −20 to +30°C; cold winters | Low–moderate: 300–600 mm; concentrated in summer | Russia; Canada; central USA |
| **Polar / Tundra** | >65° N/S | Very cold: −30 to +5°C; short cool summer | Very low: <300 mm/yr (mostly snow) | Arctic; Antarctica; northern Siberia |

## 9. Evidence of Climate Change

**Climate change:** A long-term shift in global or regional climate patterns. Since the mid-20th century, the term predominantly refers to human-caused (anthropogenic) changes, particularly global warming driven by increased greenhouse gas emissions. Global average surface temperature has risen by approximately **+1.1°C since pre-industrial levels** (IPCC Sixth Assessment Report, 2021).

| Evidence | Data/Detail | Significance |
|---|---|---|
| **Rising global temperatures** | +1.1°C above pre-industrial average (1850–1900 baseline). The 2010s were the warmest decade on record. 2023 was the hottest year ever recorded globally. | Direct measure of global warming; affects ecosystems, food production, and sea levels |
| **Glacier retreat** | Rhône Glacier (Switzerland): retreated over 3 km since 1870; lost ~20% of volume since 1900. Arctic sea ice extent has decreased by ~13% per decade since 1979. Himalayan glaciers losing mass rapidly. | Contributes to rising sea levels; threatens freshwater supply for millions |
| **Rising sea levels** | Global average sea level rising at +3.3 mm/year; total rise of ~20 cm since 1900. Accelerating: rate doubled since 1993. | Increased flood risk in low-lying coastal areas (Bangladesh, Maldives, Pacific islands, coastal UK cities) |

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| | | |
|---|---|---|
| **More frequent extreme weather** | Heatwaves more frequent and intense (UK 40°C, 2022); storms more powerful; droughts longer; flooding events more common in UK winters. | Economic damage, loss of life, food insecurity, displacement of people |
| **Earlier spring** | UK blossom dates (cherry, hawthorn) advancing by ~10–15 days since 1950s. Bird nesting dates earlier. Swallows arriving in UK earlier each decade. | Disrupts seasonal timing (phenological mismatch) between plants, insects, and birds |
| **Shifting species ranges** | UK butterfly species moving northward by c. 35–50 km per decade. Mediterranean species (e.g. little egret, bee-eater) colonising southern England. Bleaching of Great Barrier Reef (record ocean temperatures 2016, 2022, 2024). | Biodiversity loss; ecosystem disruption; extinction risk for cold-adapted species |

# 10. Causes of Climate Change

## Natural Causes

**Milankovitch cycles:** Long-term, predictable changes in Earth's orbit and orientation relative to the Sun, which alter the amount and distribution of solar energy received. Three components:

- **Eccentricity** — the shape of Earth's orbit changes from circular to more elliptical over ~100,000 years. A more elliptical orbit increases the seasonal variation in solar radiation.
- **Axial tilt (obliquity)** — the tilt of Earth's axis varies between 22.1° and 24.5° over ~41,000 years. Greater tilt = stronger seasons (hotter summers, colder winters).
- **Precession** — Earth wobbles on its axis like a spinning top, completing a cycle over ~26,000 years. This alters which hemisphere points toward the Sun during which season.

Milankovitch cycles explain the timing of ice ages but are too slow to explain rapid 20th/21st century warming.

**Volcanic eruptions:** Large volcanic eruptions inject sulphur dioxide (SO⊂2;) into the stratosphere. SO⊂2; reacts with water vapour to form sulphate aerosols, which reflect incoming solar radiation

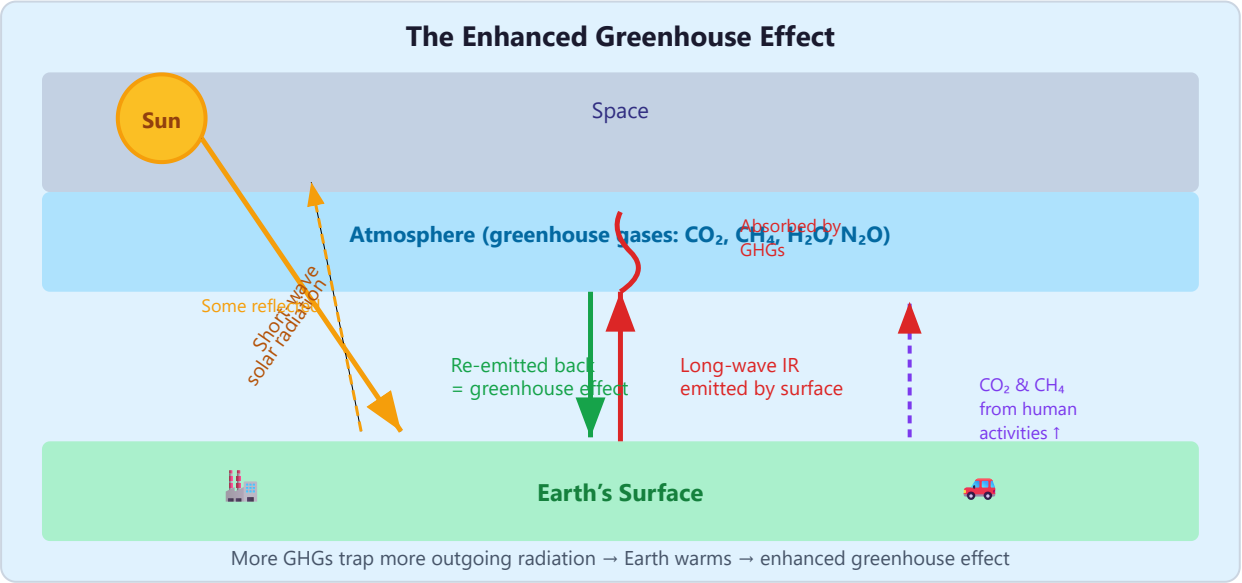
back to space, causing *short-term cooling*. Example: Mt Pinatubo eruption (Philippines, 1991) lowered global temperatures by ~0.5°C for 1–2 years. Volcanoes also emit CO⊂2;, but this is a tiny fraction of human emissions.

**Solar output variations:** Changes in the Sun's energy output (solar irradiance) affect Earth's temperature. The 11-year sunspot cycle causes minor variations. The *Maunder Minimum* (1645–1715) was a period of reduced solar activity coinciding with the Little Ice Age. However, solar output has been flat or declining slightly since 1980 while temperatures have risen — ruling out solar activity as the main cause of recent warming.

## Human Causes — The Enhanced Greenhouse Effect

**Greenhouse effect (natural):** Solar radiation passes through the atmosphere to warm Earth's surface. Earth re-emits this energy as long-wave infrared radiation. Greenhouse gases (water vapour, CO⊂2;, methane, nitrous oxide) in the atmosphere absorb some of this outgoing radiation and re-emit it back toward Earth, keeping the planet ~33°C warmer than it would otherwise be. This is a natural and essential process.

**Enhanced (anthropogenic) greenhouse effect:** Human activities have increased concentrations of greenhouse gases far beyond natural levels, trapping more heat and causing additional warming. CO⊂2; has risen from ~280 ppm (pre-industrial) to over **420 ppm** in 2024.



| Greenhouse Gas | Main Human Sources | Relative Impact |
|---|---|---|
| **Carbon dioxide (CO⊂2;)** | Burning fossil fuels (coal, oil, gas) for energy and transport; cement production; deforestation (trees absorb CO⊂2; when cut down this store is released and future absorption is lost) | Currently 420 ppm; responsible for ~64% of enhanced greenhouse effect |
| **Methane (CH⊂4;)** | Livestock farming (cattle belching/manure); rice paddies; landfill waste decomposition; natural gas leaks | ~80x more powerful than CO⊂2; over 20 years; responsible for ~17% of warming |
| **Nitrous oxide (N⊂2;O)** | Fertiliser use in agriculture; burning fossil fuels | ~300x more powerful than CO⊂2; responsible for ~6% of warming |
| **Deforestation (indirect)** | Removing forests (for farming, logging, urbanisation) releases stored carbon AND reduces future CO⊂2; absorption. 15% of global emissions come from deforestation. | Amazon deforestation increased by 77% between 2018–2021; Congo Basin and SE Asian forests also under threat |

**Exam tip:** For higher marks on causes of climate change, clearly distinguish between **natural** causes (Milankovitch cycles, volcanic activity, solar variation) and **human** causes (fossil fuel burning, deforestation, agriculture). Examiners expect you to explain the *mechanism* — don't just list sources; explain how each cause leads to warming (or cooling). For the greenhouse effect, describe the process: solar radiation in → absorbed by surface → re-emitted as infrared → absorbed by GHGs → re-emitted back to Earth → warming.

**Exam tip — 8/12-mark questions:** On Eduqas extended questions about climate change, structure your answer as: evidence → natural causes → human causes → relative significance (human causes dominate since 1950 — IPCC consensus). Use named statistics throughout.

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